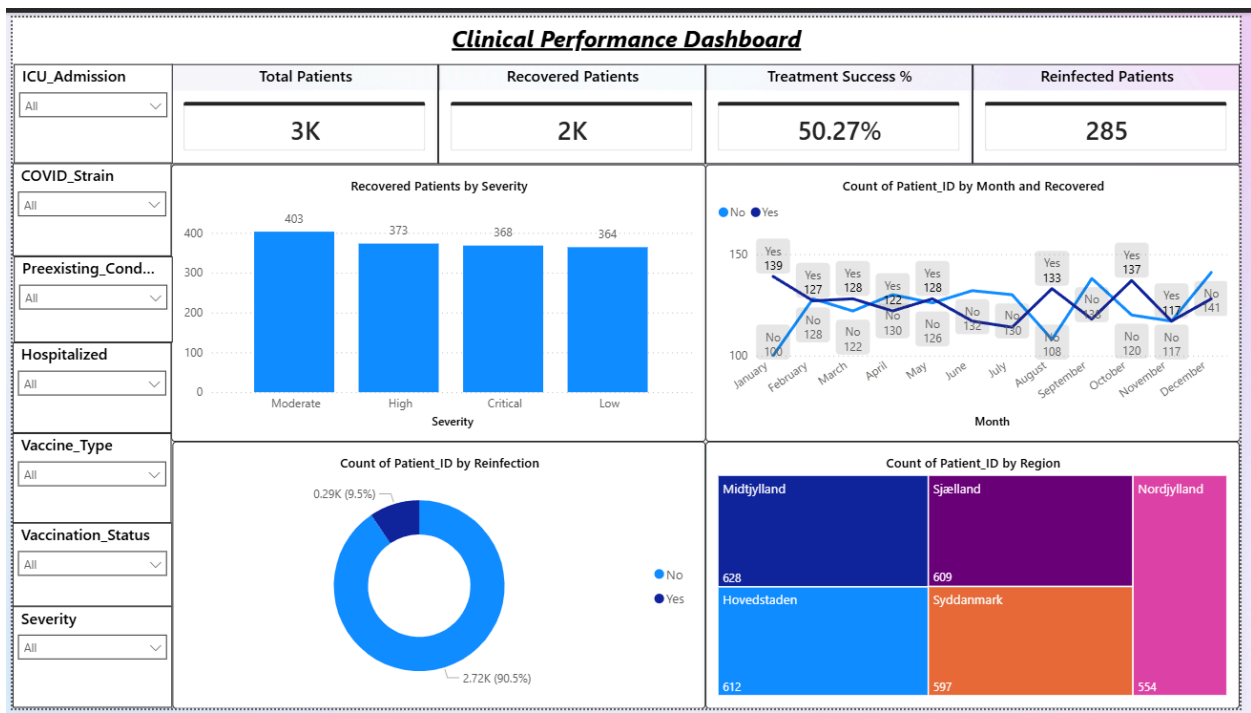


Task 14.2 Clinical Performance Dashboard Report

1. Overview

The Clinical Performance Dashboard provides a comprehensive analysis of patient outcomes, treatment effectiveness, and healthcare performance based on COVID-19 patient data. It focuses on key metrics such as recovery rates, reinfection rates, and the impact of severity, hospitalization, and vaccination on clinical outcomes.

The dashboard is designed to help stakeholders quickly assess healthcare efficiency and identify patterns influencing patient recovery.



2. Key Performance Indicators (KPIs)

Total Patients	Recovered Patients	Treatment Success %	Reinfected Patients
3K	2K	50.27%	285

Total Patients

Represents the total number of patients included in the dataset.

Value: 3K

This indicates the overall dataset size used for analysis.

Recovered Patients

Displays the number of patients who successfully recovered.

Value: 2K

This reflects the total successful treatment cases.

Treatment Success Rate

Calculated as the percentage of recovered patients out of total patients.

Value: 50.27%

This KPI measures overall treatment effectiveness.

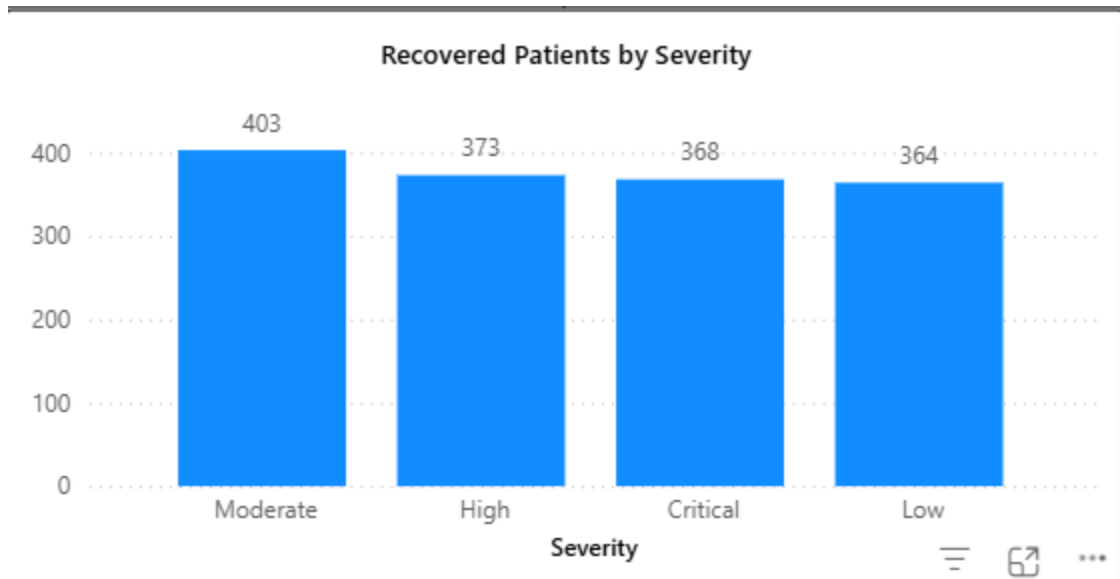
Reinfected Patients

Shows the number of patients who experienced reinfection.

Value: 285

This serves as a proxy for readmission and relapse cases.

3. Charts and Insights



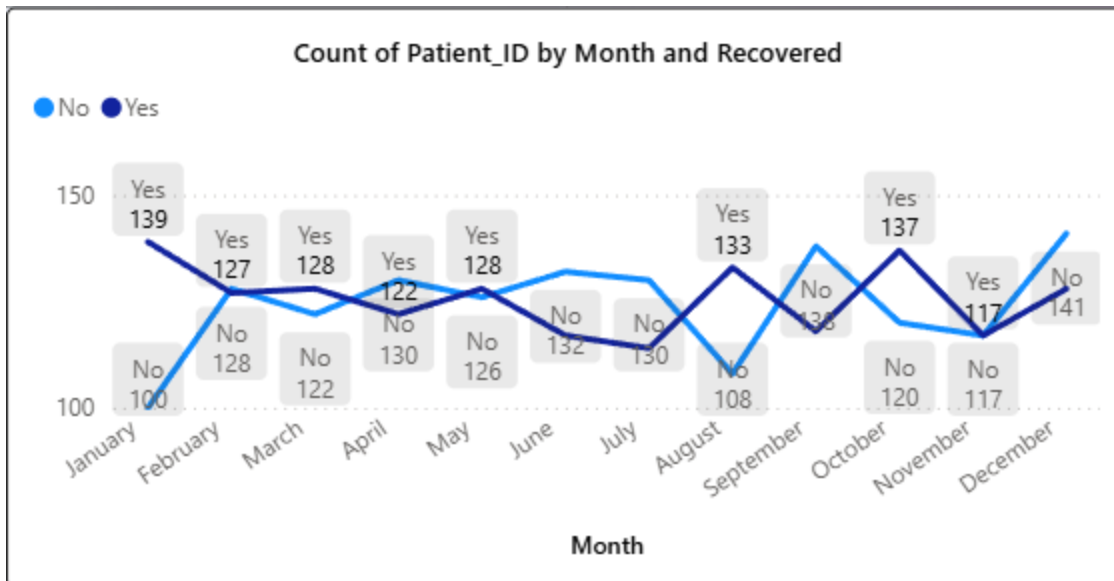
3.1 Recovered Patients by Severity (Column Chart)

This chart shows the number of recovered patients across different severity levels (Moderate, High, Critical, Low).

Insight:

- Recovery is relatively consistent across severity levels.
- Moderate cases show slightly higher recovery numbers.
- Even critical cases have comparable recovery, indicating effective treatment.

3.2 Monthly Trend of Patients and Recovery (Line Chart)

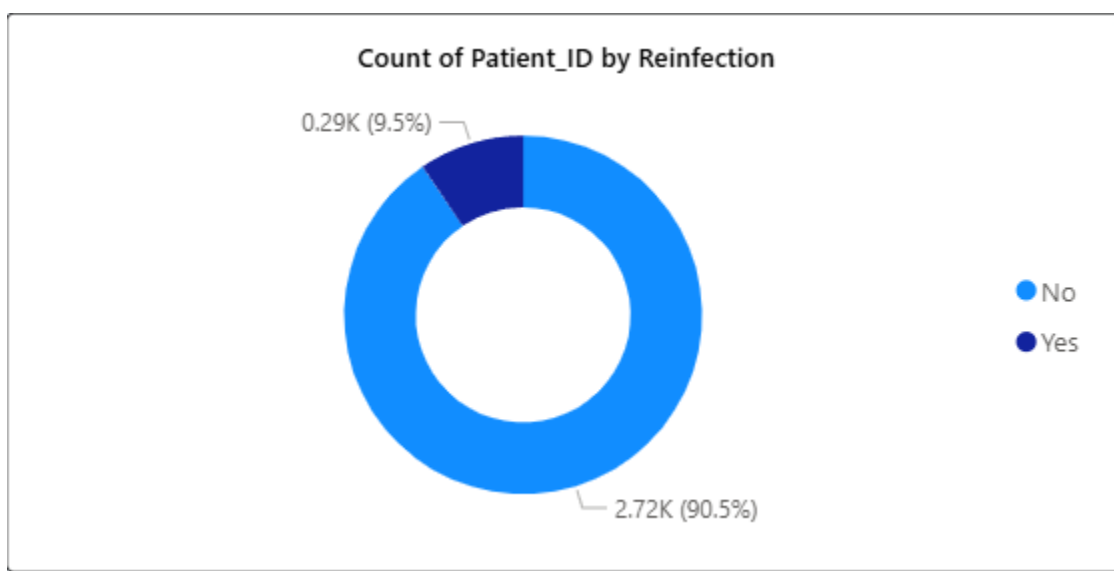


This chart tracks patient counts over months, segmented by recovery status.

Insight:

- Fluctuations indicate varying infection and recovery trends over time.
- Some months show higher recovery rates, suggesting improved treatment or lower severity cases.

3.3 Reinfection Distribution (Donut Chart)

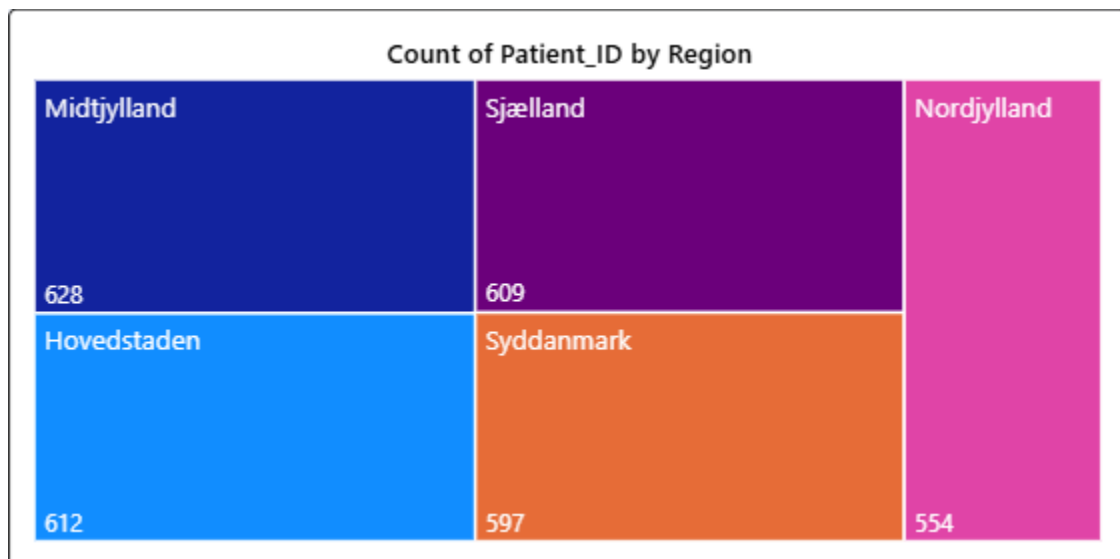


This chart displays the proportion of reinfected vs non-reinfected patients.

Insight:

- Majority of patients (~90.5%) were not reinfected.
 - A small portion (~9.5%) experienced reinfection.
 - Indicates relatively strong recovery stability.
-

3.4 Patient Distribution by Region (Treemap)

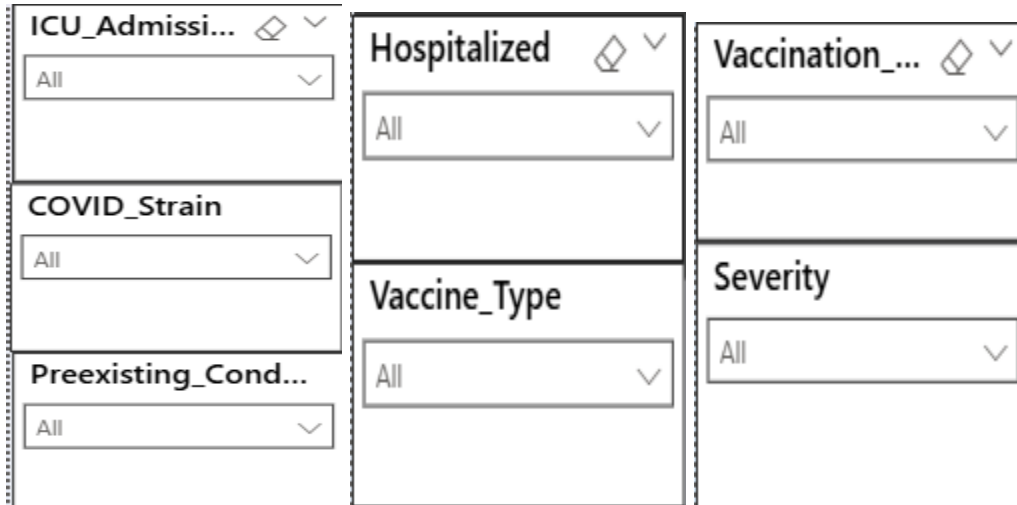


This chart visualizes patient count across different regions.

Insight:

- Patient distribution is fairly balanced across regions.
 - Slight variations indicate regional differences in case load.
 - Useful for resource allocation planning.
-

4. Slicers (Filters)



The dashboard includes multiple slicers to enable dynamic analysis:

- **ICU Admission** – Filter patients based on ICU requirement
- **COVID Strain** – Analyze impact of different virus variants
- **Preexisting Condition** – Study comorbidity effects
- **Hospitalized** – Compare hospitalized vs non-hospitalized cases
- **Vaccine Type** – Evaluate effectiveness of different vaccines
- **Vaccination Status** – Compare vaccinated vs non-vaccinated patients
- **Severity** – Focus on specific severity levels

Purpose:

These slicers allow users to drill down into specific scenarios and understand how different factors affect clinical outcomes.

5. Key Findings

- Treatment success rate is moderate at around 50%, indicating scope for improvement.
- Reinfection rate is relatively low, suggesting stable recovery for most patients.
- Severity does not drastically reduce recovery numbers, indicating effective clinical interventions.
- Regional distribution is balanced, showing no extreme concentration of cases.

- Time-based trends highlight fluctuations in infection and recovery patterns.
-

6. Conclusion

The Clinical Performance Dashboard effectively presents critical healthcare insights through KPIs, charts, and interactive filters. It enables users to evaluate treatment outcomes, identify trends, and analyze the impact of various clinical and demographic factors.

Overall, the dashboard serves as a valuable tool for monitoring healthcare performance and supporting data-driven decision-making.